

Association of Hemispheric Laterality with Neurosurgical Intervention in Intracerebral Hemorrhage: A Post-Hoc Bayesian Analysis of ATACH-2 and ERICH

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Abstract

Background: Hemispheric laterality influences clinical management in stroke. Given the increasing enthusiasm for surgical management of intracerebral hemorrhage (ICH), we evaluated the association of hemispheric ~~laterality~~lateralization with the likelihood of receiving neurosurgical intervention and subsequent 90-day functional outcomes.

Methods: This post-hoc analysis included 3477 patients with supratentorial ICH from two prospective studies: the ATACH-2 trial, conducted at 110 sites in the United States, Japan, China, Taiwan, South Korea, and Germany (2011–2015), and the ERICH study, conducted across 19 centers in the United States (2010–2015)~~studies~~. The primary outcome was neurosurgical intervention (hematoma evacuation, decompressive craniectomy, or other craniotomy). Secondary outcomes included external ventricular drain (EVD) placement, 90-day modified Rankin Scale (mRS) score, and EuroQOL domains. Hierarchical Bayesian regression models estimating the association between ICH laterality and outcomes were adjusted for age, admission Glasgow Coma Scale, ICH location, ICH volume, and intraventricular hemorrhage.

Results: ~~Of Right~~ 3477 patients, right hemispheric ICH was associated with a higher likelihood of receiving neurosurgical intervention (aOR 1.65, 95% CrI 1.28–2.14) and EVD placement (aOR 1.22, 95% CrI 1.04–1.44) compared to left hemispheric ICH. This association was directionally consistent across ICH locations, study cohorts, and prior specifications. Despite the lower rates of ~~aggressive surgical~~ intervention management, in patients with left hemispheric ICH, there demonstrated was similar no meaningful difference in 90-day functional outcomes; in exhibiting patient equivalent with mRS left scores and right potentially hemispheric better ICH outcomes in specific health-related quality of life domains.

Conclusions: Right hemispheric ICH is associated with a higher likelihood of neurosurgical intervention compared to left hemispheric ICH, though 90-day functional outcomes remain similar between groups. While these findings highlight a potential laterality-~~associated~~driven ~~variation~~disparity in ~~neurosurgical treatment intervention~~paradigms, residual confounding from unmeasured anatomical severity and family preferences cannot be excluded. Prospective studies are needed to delineate the explicit clinical and prognostic drivers of surgical decision-making to optimize treatment allocation in ICH.

Trial Registration: Clinicaltrials.gov Identifier: ATACH-2, NCT01176565; ERICH, NCT01202864

Introduction

Intracerebral hemorrhage (ICH) remains a challenging disease, marked by its sudden onset, morbidity, and mortality.^{1,2} While management of ischemic stroke has advanced considerably over the last decade, the acute management of ICH remains limited to blood pressure control, coagulopathy reversal, and perhaps surgical evacuation of hemorrhage.^{3–5,4} These efforts have focused on reducing mortality and preventing additional morbidity due to secondary injury.

Clinical presentations for patients with stroke are varied based on the hemisphere affected, where hemispheric dominance dictates cognitive processing, attention, language, and emotional responses. Consequently, when confronted with ICH, the specific hemisphere affected can profoundly influence not only the clinical presentation but also management decisions and prognostic outcomes.

Previous research involving patients with ischemic stroke ~~provides~~^{provide} evidence that those with right hemispheric injury present later to the hospital and may not be treated as frequently with thrombolytic therapy.^{65–98} Some practitioners hesitate to offer decompressive craniectomy in patients with left dominant-hemispheric ischemic infarcts despite literature demonstrating that long-term functional outcomes and health-related quality of life (HRQoL) ~~are~~^{appear} ~~comparable to~~ ~~be equivalent~~ for patients with left hemispheric injuries.^{109–1312}

Amidst an increasing enthusiasm for surgical management of ICH^{3,4,1413}, it is important to understand ~~whether~~^{if} hemispheric ~~laterality~~^{lateralization} is associated with differences in surgical decision making, as has been observed for ischemic stroke. We aimed to evaluate the

association of hemispheric ~~laterality~~^{lateralization} of ICH with the likelihood of receiving neurosurgical intervention as well as with functional outcomes.

Methods

Study Design and Participants

We conducted a post-hoc analysis using data from two prospective studies: the Antihypertensive Treatment of Acute Cerebral Hemorrhage II (ATACH-2) trial and the Ethnic/Racial Variations of Intracerebral Hemorrhage (ERICH) study. ATACH-2 was a large, multicenter, international, open-label, blinded-endpoint randomized controlled trial that enrolled 1000 patients with ICH ~~across from 110 sites to~~^{in 2016} the United States, Japan, China, Taiwan, South Korea, and Germany from 2011-2015 and randomized them to either intensive blood pressure control (110-139 mm Hg) or standard blood pressure control (140-179 mm Hg). ERICH was a large, multicenter, prospective ~~case-control~~^{observational} study ~~conducted~~^{conducted} across 19 centers in the United States on patients admitted with ICH from 2010-~~2015 to~~²⁰¹⁷, with particular emphasis on recruitment of a racially and ethnically diverse cohort. Only ICH cases were included in the present analysis. The protocols and results of both studies have been previously published elsewhere¹⁵⁻¹⁷~~elsewhere~~^{14,15}.

We included all patients enrolled in these two studies with supratentorial location of ICH. We excluded patients with cerebellar or brainstem ICH as well as any patient with primary intraventricular hemorrhage (IVH).

The study was approved by the institutional review board of the Wake Forest School of Medicine (IRB00098198).

Exposures and Outcomes

Our exposure was hemispheric laterality (left versus right). Our primary outcome was neurosurgical intervention as a binary variable defined as receiving a hematoma evacuation, decompressive craniectomy, or other craniotomy.

To explore differences in other aspects of aggressive care, we defined several secondary outcomes: receipt of external ventricular drain (EVD), receipt of tracheostomy, days of mechanical ventilation, receipt of do-not-resuscitate (DNR) order at any point in the hospitalization, withdrawal of life-sustaining therapy (WLST) at any point in the hospitalization, and early WLST (defined as < 72 hours from symptom onset). Receipt of an EVD was considered as a distinct outcome from neurosurgical intervention because the decision-making for each is slightly different. Placement of an EVD involves consideration of the presence of hydrocephalus and IVH, while neurosurgical intervention is affected by hematoma location, volume, midline shift, and other factors.

Our primary functional outcome was score on the modified Rankin scale (mRS) at 90 days analyzed via ordinal regression. We also examined quality of life as assessed by the European Quality of Life-5 dimensions (EQ-5D) questionnaire. In both studies, the mRS was obtained via a standardized, structured interview and the EQ-5D via a standardized questionnaire.

Statistical Analysis

Associations between hemispheric laterality and clinical outcomes were evaluated using hierarchical Bayesian regression models, adjusting for the covariates listed below with a random intercept applied to the enrolling study cohort.

We employed Bayesian statistical methods for this analysis due to several advantages over traditional frequentist approaches. Bayesian methods allow for the incorporation of prior knowledge or beliefs into the analysis, which can be particularly useful in observational studies where prior information about relationships between variables may exist. This is achieved through the use of prior distributions, which are combined with the observed data to produce posterior distributions representing updated beliefs about the parameters of interest. Furthermore, Bayesian methods provide a natural framework for quantifying uncertainty through credible intervals (CrI), which offer a more intuitive interpretation than frequentist confidence intervals. Finally, Bayesian modeling facilitates the estimation of posterior probabilities, allowing for direct probabilistic statements about hypotheses, such as the probability that an effect size (e.g., odds ratio [OR] or incidence rate ratio [IRR]) is greater than 1, which is often of greater clinical relevance than a p-value.^{1846,1947} This report adheres to the Bayesian Analysis Reporting [and STROBE](#) Guidelines.^{20,2148}

Binary, ordinal, and count-based outcomes (~~days-of-mechanical-ventilation~~) were modeled using hierarchical Bayesian logistic, ordinal logistic, and zero-inflated negative binomial regressions, respectively. The European Quality of Life Visual Analog Scale (EQ-VAS) was scaled to a proportion and analyzed via hierarchical Bayesian zero-one inflated beta (ZOIB) regression. Because our adjusted models included an interaction term between hemispheric laterality and ICH location, the raw main-effect coefficients for laterality are conditional on the reference location. To calculate the overall average effect of laterality across the entire cohort, we applied marginal standardization (G-computation) to all models. This allowed us to estimate average

marginal effects while averaging over the empirical distribution of all covariates and the random intercept.

To account for potential clustering of clinical decision-making by hospital, we ideally would have included a random intercept for the study site in all models. However, site-level identifiers were not available for the ERICH cohort. To assess the potential impact of omitting site-level clustering, we performed a sensitivity analysis restricted to the ATACH-2 cohort, where site identifiers were available. We fitted the primary model for neurosurgical intervention with and without a random intercept for hospital site and compared the posterior estimates for the association with laterality.

All analyses were performed in R version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria). All packages used for the analysis are listed in the [SupplementalSupplementary Appendix](#).

Covariate Selection

We constructed directed acyclic graphs (DAGs) to model our causal assumptions and to guide covariate selection for each estimand.^{2219–2421} For neurosurgical intervention and other acute care outcomes, the target estimand was the total effect of hemispheric laterality on treatment decisions. Models were adjusted for baseline confounders that independently influence both the clinical presentation associated with hemispheric laterality and the likelihood of intervention: age, admission Glasgow Coma Scale (GCS) score, ICH location (categorized as lobar, thalamic, or basal ganglia), baseline ICH volume, and presence of intraventricular hemorrhage (IVH).

For functional outcomes (mRS and EQ-5D), the target estimand was the total effect of hemispheric laterality, encompassing all causal pathways including those mediated through neurosurgical intervention and other post-baseline care decisions. These models were adjusted for the same baseline confounders but did not condition on neurosurgical intervention, as it lies on the causal pathway from laterality to functional outcome; adjusting for a mediator would block the clinically relevant pathway and risk introducing collider stratification bias.²³²⁹ The DAGs and further details of the covariate selection rationale are presented in the [SupplementalSupplementary](#) Appendix.

Prior Distribution Specification

To assess the robustness of our findings, we applied four sets of prior distributions for the primary exposure across all models. Because the models utilize different link functions depending on the outcome distribution (e.g., logit for binary/ordinal outcomes, log for count outcomes), these priors correspond to log-odds ratios (~~OR~~), log-incidence rate ratios (~~IRR~~), or log-mean differences.

The four prior sets were: (1) a neutral, weakly informative prior centered on a null effect (Normal(0, 0.5)); (2) an informative left-hemisphere prior favoring outcomes in left-hemispheric patients (Normal(-0.22, 0.175)); (3) an informative right-hemisphere prior favoring outcomes in right-hemispheric patients (Normal(0.18, 0.175)); and (4) a vague, diffuse prior (Normal(0, 5)) that provides minimal information to the posterior, approximating the assumptions of a frequentist analysis. Further rationale and empirical evaluations of these priors are detailed in the [SupplementalSupplementary](#) Appendix.

Computation and Description of the Posterior Distribution

For the effect of hemispheric laterality, we report the following metrics derived from the marginalized posterior distributions: 1) the median posterior effect size with 95% ~~credible intervals~~ (CrI); 2) the posterior probability of any directional difference; 3) the posterior probability of a substantial difference, defined a priori; and 4) the percentage of the posterior distribution falling ~~within~~outside a predefined region of practical equivalence (ROPE).

Effect sizes are reported as marginal odds ratios (OR) for binary and ordinal models, marginal incidence rate ratios (IRR) for the count component of the zero-inflated negative binomial ~~model~~models, and average marginal expected mean differences on the original 0–100 scale for the EQ-VAS (derived via G-computation). We defined a substantial difference as an OR~~odds or ratio~~ IRR(OR) > 1.2, (or a mean difference exceeding~~of~~ 5 points for the EQ-Euro~~-VAS~~). In the context of invasive procedures such as decompressive~~hemieraniectomy~~ craniectomy or hematoma evacuation, we posited that a 20% relative shift in treatment odds represents a minimum threshold that would meaningfully impact clinical and shared decision-making. ~~The full rationale for these thresholds and the ROPE bounds for ORs and IRRs are detailed in the Supplementary Appendix.~~

The ROPE defines the range of effects we considered practically indistinguishable from no effect.²⁰ For odds and incidence rate ratios we set this at 0.95 to 1.05, corresponding to a ±5% relative change in the odds or rate — a shift too small to plausibly alter triage, counseling, or shared decision-making. For the EQ-VAS we used ±2 points on the 0–100 scale, chosen to be conservatively narrow. Throughout the manuscript and supplement, ROPE values are reported as the percentage of the posterior distribution falling within the region of practical equivalence.

Missing Data Treatment and Sensitivity Analyses

Baseline covariates used in model adjustment (e.g., age, GCS, ICH volume) had minimal missingness (< 1%). Missingness for 90-day functional outcomes was more substantial but was balanced across laterality groups. Detailed missingness rates, assessments supporting the Missing at Random (MAR) assumption, and imputation diagnostics are reported in the [SupplementalSupplementary](#) Appendix.

Our primary analysis used multiple imputation by chained equations (MICE) to address missing data, generating 20 imputed datasets using the mice package. Imputation models included all analysis covariates and all primary and secondary outcomes. Including all functional outcomes (e.g., mRS and EuroQOL domains) in the imputation equations allowed them to serve as highly informative auxiliary variables for one another, optimizing the imputation of missing outcome data. Predictive mean matching was used for continuous and ordinal variables, logistic regression for binary variables, and polytomous regression for nominal variables. Each Bayesian model was fitted separately to each imputed dataset, and posterior draws were pooled across imputations to produce the reported estimates and CrI. A complete-case sensitivity analysis was also performed.

In addition to the missing data analysis, we analyzed our models using the four sets of prior distributions described above. To evaluate potential effect modification on acute care decisions, we fitted models containing formal interaction terms between hemispheric laterality and ICH location, and laterality and study source. Because missingness in baseline covariates and acute surgical outcomes was negligible, these exploratory subgroup interaction models—as well as a prespecified sensitivity analysis for all outcomes—were fitted to complete-case data (excluding patients missing the specific outcome being modeled) for computational efficiency. Subgroup-

specific aORs and interaction Ratio of Odds Ratios (ROR) with 95% CrI were computed to evaluate heterogeneity.

Data Availability Statement

The ATACH-2 (PI: Adnan I. Qureshi, MD, NCT01176565) and ERICH (PI: Daniel Woo, MD, MS, NCT01202864) data and analyses presented in this manuscript are based on the research files downloaded from the Archived Clinical Research web site of the National Institute of ~~Neurological~~~~Neurologic~~ DisordersDisease and Stroke (NINDS). The datasets can be obtained after approval of a written proposal as outlined on the NINDS website. As outlined in the NINDS Data Request Agreement, the analysis code is publicly available at <https://github.com/nikhilpatelmd/ich-laterality>.

Results

Baseline Characteristics

A total of 3477 patients were included in the analysis, of which 1720 (49%) had left hemispheric ICH (Table 1). They had ~~higher National Institutes of Health Stroke Scale (NIHSS) scores,~~ lower GCS scores, and ~~higherlower NIHSS rates scores of hypertension~~ as compared to those with right hemispheric ICH. There were 2487 (72%) patients in the cohort from ERICH and 990 (28%) from ATACH-2.

Hemispheric Laterality and Neurosurgical Intervention

Associations between hemispheric laterality and primary and secondary outcomes are shown in Table 2. 86 (5.0%) patients with left hemispheric ICH underwent neurosurgical intervention, compared to 144 (8.2%) with right hemispheric ICH (aOR 1.65, 95% CrI 1.28–2.14). The

posterior probability that the aOR exceeded 1 was > 99.9%, and the posterior probability of a substantial difference (aOR > 1.2) was 99.3%. ~~Less~~^{Fewer} than ~~<~~^{<=} 0.1% of the posterior distribution fell within the ROPE.

In exploratory subgroup analyses, the association between right hemispheric ICH and neurosurgical intervention remained directionally consistent across all evaluated ICH locations and source studies. Formal interaction analyses yielded wide credible intervals for the Ratio of Odds Ratios (ROR), providing insufficient evidence to conclude that the laterality effect differed meaningfully across these subgroups (Supplemental Table 10).

Figure 1 displays the covariate-adjusted marginal predicted probabilities of neurosurgical intervention and the prior sensitivity analysis. Panel A shows the posterior distributions of covariate-adjusted absolute predicted probabilities of neurosurgical intervention for left and right hemispheric ICH under the neutral prior (left hemisphere: 4.86%, 95% CrI 3.98–5.84%; right hemisphere: 8.04%, 95% CrI 6.93–9.23%). Panel B shows the posterior distributions of the absolute ~~posterior~~ probability difference (Right – Left) across all four prior specifications. Under the neutral prior, the median ~~posterior~~ probability difference was 3.18 percentage points (95% CrI 1.75–4.62), with P(Right > Left) exceeding 99.9%. Corresponding estimates were 2.21 percentage points (95% CrI 0.97–3.45) under the left-hemisphere prior, 2.84 percentage points (95% CrI 1.60–4.09) under the right-hemisphere prior, and 3.38 percentage points (95% CrI 1.90–4.89) under the diffuse prior, with P(Right > Left) exceeding 99.9% under all specifications. Further details are available in Supplemental Tables ~~1114-1317~~.

In a sensitivity analysis restricted to the ATACH-2 cohort, a random intercept for the enrolling hospital was included to account for site-level clustering. The base model yielded an adjusted odds ratio of 2.23 (95% CrI: 1.31–3.76). With the inclusion of the site-level random effect, the adjusted odds ratio was 1.86 (95% CrI: 1.10–3.16). For this hierarchical model, the posterior probability ~~that of the any directional effect (aOR exceeded > 1)~~ was 99%, the posterior probability of an aOR > 1.2 was ~~94.795%~~, and ~~1.2%~~ of the posterior distribution fell within the region of practical equivalence (~~Supplemental~~Supplementary Table ~~1948~~).

Hemispheric Laterality and Secondary Outcomes

A total of 236 (13.7%) with left hemispheric ICH had an EVD placed as compared to 272 (15.5%) with right hemispheric ICH (aOR 1.22, 95% CrI 1.04–1.44). The posterior probability that the aOR exceeded 1 was 99.2%, and the posterior probability of a substantial difference (aOR > 1.2) was 58.2%. 3.3% of the posterior distribution fell within the ROPE.

Median days on mechanical ventilation were 6 (3 – 12) and 6 (3 – 9) for left and right hemispheric ICH, respectively (IRR 0.99, 95% CrI 0.89–1.09). The posterior probability that the IRR exceeded 1 was 39%, and 65.6% of the posterior distribution fell within the ROPE.

Rates of tracheostomy were similar between groups (19 (1.1%) left vs. 19 (1.1%) right; aOR 1.13 (0.66 - 1.91)), with a posterior probability that the aOR exceeded 1 of 67.2% and 13.1% of the posterior distribution within the ROPE. Rates of WLST (152 (8.8%) left vs. 143 (8.1%) right; aOR 0.95 (0.76 - 1.17)), early WLST (45 (2.6%) left vs. 39 (2.2%) right; aOR 0.93 (0.63 - 1.35)), and DNR orders (195 (11.3%) left vs. 202 (11.5%) right; aOR 1.01 (0.85 - 1.21)) were also similar across hemispheres, with P(aOR > 1) of 30.2%, 34.5%, and 55.2%, respectively.

Hemispheric Laterality and Functional Outcomes

Associations between hemispheric laterality and functional outcomes are shown in Table 3. The aOR for a worse mRS score at 90 days for right hemispheric ICH compared to left hemispheric ICH was 0.99 (95% CrI 0.98–1.00), with a posterior probability that the aOR exceeded 1 of 0.7% and > 99.9% of the posterior distribution within the ROPE. Figure 23 displays the covariate-adjusted absolute marginal ~~posterior~~~~predicted~~ ~~probability~~~~cumulative~~ distributions of mRS at 90 days for left and right hemispheric ICH under the neutral prior.

For Mobility and Self-Care, the posterior probability that the aOR exceeded 1 was > 99.9% for both domains (aOR 1.03, 95% CrI 1.01–1.05 for Mobility; aOR 1.03 (1.02 - 1.05) for Self-Care), with a posterior probability of a substantial difference (aOR > 1.2) of < 0.1% for both, and 98.3% and 96.4% of the posterior distributions within the ROPE, respectively. For Pain/Discomfort, the aOR was 1.06 (1.01 - 1.11) with P(aOR > 1) of 99.1% and 38.3% of the posterior distribution within the ROPE. For Anxiety/Depression, the aOR was 1.04 (0.99 - 1.09) with P(aOR > 1) of 95.3% and 64% of the posterior distribution within the ROPE. For Usual Activities, the aOR was 0.99 (0.99 - 1.00) with P(aOR > 1) of 3.1% and > 99.9% of the posterior distribution within the ROPE. For the ~~EQ-Euro~~-VAS, the estimated mean difference was -1.7 points (95% CrI -3.6–0.2), with a posterior probability of 96.1% that right hemispheric ICH was associated with a lower score, 62.3% of the posterior distribution within the ROPE, and a posterior probability of a substantial difference (mean difference < -5 points) of < 0.1%.

Results from sensitivity analyses using alternative priors and complete case analysis are presented in Supplemental Tables ~~11-14~~~~16~~, 18 and are directionally consistent with these results.

Discussion

In this pooled analysis of 3477 patients with ICH, we observed an association between right hemispheric ICH and a higher likelihood of receiving neurosurgical intervention compared to left hemispheric ICH. Specifically, patients with right hemispheric ICH had 1.65 times the adjusted odds of receiving a neurosurgical intervention and 1.22 times the adjusted odds of receiving an EVD, although this was a more modest association. This finding association was directionally consistent across ICH locations, study cohorts, and prior specifications, though incorporating a hospital-level random intercept to account for site-level clustering modestly attenuated the effect. Despite lowerless rates aggressive of surgical intervention management, in patients with left hemispheric ICH, there demonstrated was similar no meaningful difference in 90-day functional outcomes in and patients potentially with better left quality and of right life hemispheric ICH measures. This finding equivalence aligns with previous literature in both ICH and ischemic stroke, where a clear association between hemispheric laterality and functional outcomes has often been difficult to establish.^{9,10,11,42,13,22,25,26,23}

The exact mechanisms driving these associations remain unclear, and the observational nature of the data precludes causal inferences. In both the ATACH-2 and ERICH studies, neurosurgical candidacy was not governed by strict trial protocols but was left entirely to clinician discretion. Furthermore, ATACH-2 excluded patients who required immediate, life-saving hematoma evacuation upon presentation. Therefore, The the observed association disparity in our cohort likely reflects a complex interplay of clinical triage, unmeasured anatomical severity, and shared decision-making rather than isolated treatment bias.

One plausible explanation for this associationdisparity involves language-related prognostic framing. Left-hemispheric ICH is frequently associated with aphasia, a deficit that clinicians, patients, and families historically weigh heavily when anticipating long-term quality of life.²⁷²⁴
²⁹²⁶ Prognostic framing that emphasizes the risk of severe aphasia may heavily influence surrogate decision-makers during early goals-of-care discussions, potentially driving lower intervention rates. This dynamic is consistent with the well-characterized self-fulfilling prophecy in neurocritical care. Conversely, severe neglect—which commonly complicates right-hemispheric lesions and profoundly impairs mobility and self-care—may be less frequently emphasized as a barrier to surgical candidacy, despite its distinct impact on caregiver burden and long-term disability.

This study has several important limitations. First, we lacked granular data on variables ~~critical~~ that ~~on~~ founders shape surgical decision-making, such as specific mass effect ~~trajectories~~, degree of midline shift, hydrocephalus, and explicit family preferences. These unmeasured factors ~~clinical nuances are primary drivers of surgical decision-making and~~ could independently influence both perceived surgical candidacy and functional outcomes. Second, we relied on hemispheric laterality as an imperfect proxy for true language dominance; because handedness data were unavailable, the expected 5–10% of patients with atypical dominance likely introduced non-differential misclassification, biasing our estimates toward the null. Third, absolute surgical event rates in this cohort were relatively low. Consequently, it is unclear if these associations are transportable to contemporary, unselected ICH populations or to patients with moderate-to-severe ICH who possess immediate indications for emergent surgery. Finally, attitudes toward disability and the frameworks used for shared decision-making carry significant cultural nuance,

and our findings—derived primarily from cohorts in the United States—may not generalize across diverse global and socioeconomic settings.

With recent randomized trials demonstrating the benefit of hematoma evacuation and decompressive craniectomy in appropriately selected ICH patients^{3,4}, understanding the specific drivers of discretionary surgical decisions is increasingly critical. Future research should prospectively investigate the precise clinical criteria used to determine surgical candidacy, while also exploring the details of prognostic framing and shared decision-making across varying cultural contexts to ensure equitable and evidence-based neurosurgical care.

Conclusions

In this post-hoc observational analysis, right hemispheric ICH was associated with a higher likelihood of neurosurgical intervention compared to left hemispheric ICH. However, 90-day functional outcomes were similar across hemispheres. While these findings highlight a ~~potential-~~ laterality-~~associated~~driven variation~~disparity~~ in neurosurgical treatment intervention~~paradigms~~, residual confounding from unmeasured anatomical severity and family preferences cannot be excluded. Future prospective studies are needed to delineate the explicit clinical and prognostic drivers of surgical decision-making to optimize treatment allocation in ICH.

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Tables

Table 1: Demographic and Clinical Characteristics of Patients with Left and Right Hemispheric Intracerebral Hemorrhage at Baseline

Characteristic	Overall N = 3,477	Left N = 1,720	Right N = 1,757	SMD
Age	61 (52, 73)	61 (52, 72)	61 (52, 73)	-0.01
Sex				0.05
Female	1,396 (40%)	713 (41%)	683 (39%)	
Male	2,081 (60%)	1,007 (59%)	1,074 (61%)	
Race				0.06
American Indian or Alaskan Native	7 (0.2%)	5 (0.3%)	2 (0.1%)	
Asian	557 (16%)	263 (15%)	294 (17%)	
Black or African-American	1,022 (29%)	510 (30%)	512 (29%)	
Other/Not Reported	16 (0.5%)	7 (0.4%)	9 (0.5%)	
White	1,875 (54%)	935 (54%)	940 (54%)	
Ethnicity				0.04
Hispanic or Latino	129 (3.7%)	70 (4.1%)	59 (3.4%)	
Not Hispanic or Latino	2,296 (66%)	1,141 (66%)	1,155 (66%)	
Unknown/Not Reported	1,052 (30%)	509 (30%)	543 (31%)	
LNW to ED Presentation (mins)	78 (60, 180)	74 (60, 180)	83 (59, 180)	-0.06
Baseline SBP (mm Hg)	190 (165, 215)	190 (165, 216)	190 (165, 214)	0.02
Baseline NIHSS	11 (6, 16)	11 (6, 17)	11 (7, 14)	0.24
Baseline GCS	15 (12, 15)	14 (11, 15)	15 (13, 15)	-0.26
Hypertension	784 (23%)	366 (21%)	418 (24%)	0.06
Type II Diabetes	705 (20%)	348 (20%)	357 (20%)	0.00
Previous Stroke	583 (17%)	293 (17%)	290 (17%)	0.01
Tobacco Use	1,622 (47%)	780 (45%)	842 (48%)	0.05
ICH Location				0.12
Basal Ganglia	1,304 (38%)	688 (40%)	616 (35%)	
Lobar	1,016 (29%)	505 (29%)	511 (29%)	
Thalamus	1,157 (33%)	527 (31%)	630 (36%)	
ICH Volume (mL)	11 (5, 26)	11 (5, 24)	12 (5, 27)	-0.06
IVH	1,304 (38%)	624 (36%)	680 (39%)	0.05

Study				0.05
ATACH-2	990 (28%)	472 (27%)	518 (29%)	
ERICH	2,487 (72%)	1,248 (73%)	1,239 (71%)	

Values are n/total n (%) or median (25th–75th percentiles) [no. of available patients in case of missing data]. LNW indicates last known well; ED, emergency department; SBP, systolic blood pressure; NIHSS, National Institutes of Health Stroke Scale; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IVH, intraventricular hemorrhage; and SMD, standardized mean difference.

Table 2: Association of Hemispheric Laterality with Primary and Secondary Outcomes

<u>Outcome</u>	<u>Left Hemisphere</u>	<u>Right Hemisphere</u>	<u>aOR / IRR (95% CrI)</u>	<u>Posterior Probability of difference (Est > 1)</u>	<u>Posterior Probability of substantial difference (Est > 1.2)</u>	<u>Percentage of posterior within ROPE</u>
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.65 (1.28 - 2.14)	> 99.9%	99.3%	< 0.1%
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.22 (1.04 - 1.44)	99.2%	58.2%	3.3%
Days of Mechanical Ventilation	6 (3 – 12)	6 (3 – 9)	0.99 (0.89 - 1.09)	39%	< 0.1%	65.6%
Tracheostomy	19 (1.1%)	19 (1.1%)	1.13 (0.66 - 1.91)	67.2%	41.2%	13.1%
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	0.95 (0.76 - 1.17)	30.2%	1.3%	31.5%
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	0.93 (0.63 - 1.35)	34.5%	8.8%	19.1%
DNR Order	195 (11.3%)	202 (11.5%)	1.01 (0.85 - 1.21)	55.2%	3%	41.3%

Values represent Average Marginal Effects (Odds Ratios for binary outcomes, Incidence Rate Ratios for counts). Models are adjusted for age, admission GCS, ICH location (interaction), ICH volume, IVH, and study. Days of mechanical ventilation represent median (IQR) days (data from ATACH-2 only). ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IQR, interquartile range; and IVH, intraventricular hemorrhage.

Table 2: Association of Hemispheric Laterality with Primary and Secondary Outcomes

Outcome	Left Hemisphere	Right Hemisphere	aOR / IRR (95% CI)	Probability of difference (Est > 1)	Probability of substantial difference (Est > 1.2)	ROPE
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.65 (1.28 – 2.14)	>99.9%	99.3%	<0.1%
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Table 3: Association of Hemispheric Laterality with Functional Outcomes

<u>Outcome</u>	<u>aOR / Mean Diff</u> <u>(95% CrI)</u>	<u>Posterior</u> <u>Probability of any</u> <u>difference</u>	<u>Posterior</u> <u>Probability of a</u> <u>substantial</u> <u>difference</u>	<u>Percentage of</u> <u>posterior within</u> <u>ROPE</u>
Modified Rankin Score	0.99 (0.98 - 1.00)	0.7%	< 0.1%	> 99.9%
EuroQOL - Mobility	1.03 (1.01 - 1.05)	> 99.9%	< 0.1%	98.3%
EuroQOL - Self-Care	1.03 (1.02 - 1.05)	> 99.9%	< 0.1%	96.4%
EuroQOL - Usual Activities	0.99 (0.99 - 1.00)	3.1%	< 0.1%	> 99.9%
EuroQOL - Pain/Discomfort	1.06 (1.01 - 1.11)	99.1%	< 0.1%	38.3%
EuroQOL - Anxiety/Depression	1.04 (0.99 - 1.09)	95.3%	< 0.1%	64%
EQ-VAS	-1.7 (-3.6 - 0.2)	96.1%	< 0.1%	62.3%

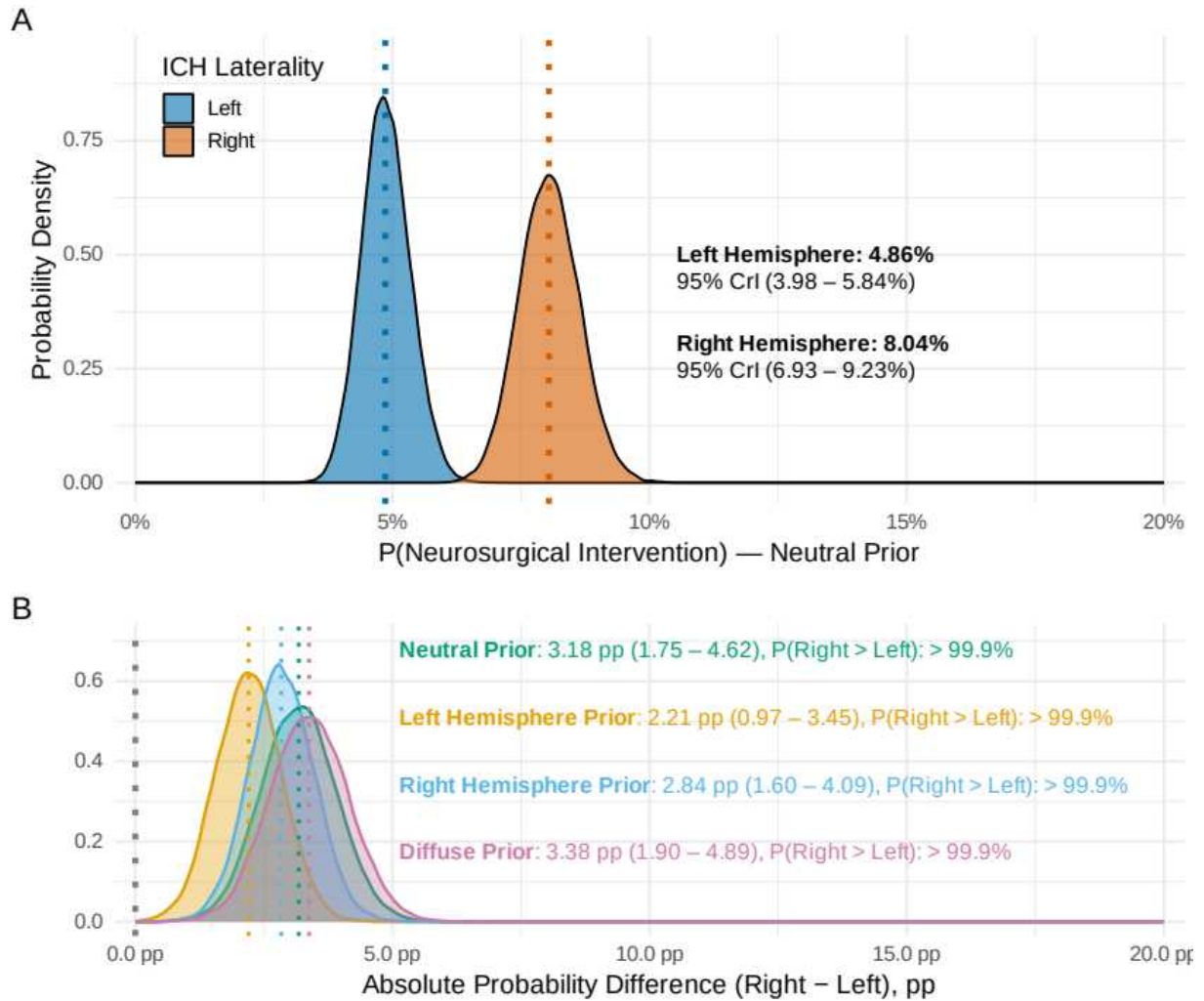
Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for EQ-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for EQ-VAS). For EQ-VAS, probability of difference is defined as Mean Difference < 0, and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Table 3: Association of Hemispheric Laterality with Functional Outcomes

Outcome	aOR / Mean Diff (95% CI)	Probability of any difference	Probability of a substantial difference	ROPE
Modified Rankin Score	0.99 (0.98–1.00)	0.7%	<0.1%	>99.9%
EuroQOL—Mobility	1.03 (1.01–1.05)	>99.9%	<0.1%	98.3%
EuroQOL—Self-Care	1.03 (1.02–1.05)	>99.9%	<0.1%	96.4%
EuroQOL—Usual Activities	0.99 (0.99–1.00)	3.1%	<0.1%	>99.9%
EuroQOL—Pain/Discomfort	1.06 (1.01–1.11)	99.1%	<0.1%	38.3%
EuroQOL—Anxiety/Depression	1.04 (0.99–1.09)	95.3%	<0.1%	64%
Euro-VAS	-1.7 (-3.6–0.2)	96.1%	<0.1%	62.3%

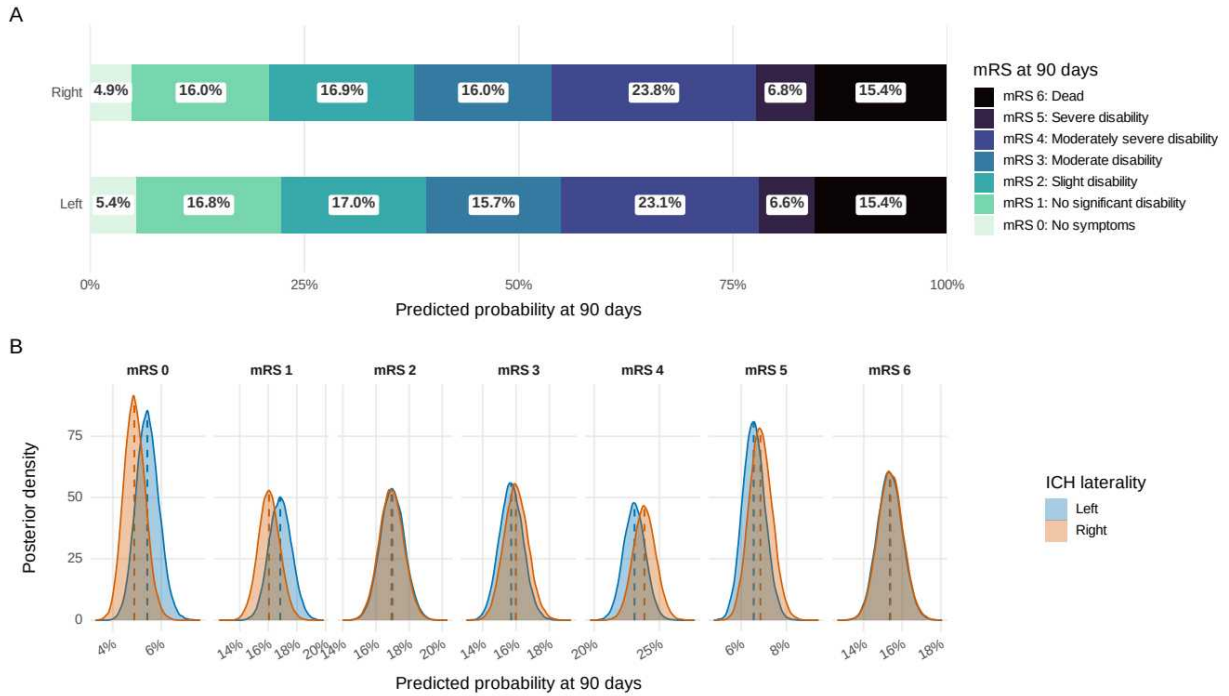
Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for Euro-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for Euro-VAS). For Euro-VAS, probability of difference is defined as Mean Difference < 0 , and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Figures



Panel A: Posterior distributions of covariate-adjusted average predicted probabilities of neurosurgical intervention by ICH hemisphere under the neutral prior. Panel B: Overlaid posterior distributions of the absolute probability difference (Right – Left hemisphere) across four prior specifications (neutral, left-hemisphere, right-hemisphere, and diffuse). All four priors are directionally consistent across prior specifications. Models were adjusted for ICH location, age, admission Glasgow Coma Scale score, ICH volume, intraventricular hemorrhage, an ICH laterality-by-location interaction, and a random intercept for enrolling study. Dotted lines indicate posterior medians. pp = percentage points. CrI = credible interval.

Figure 1: Prior Sensitivity Analysis for Neurosurgical Intervention



Panel A: Covariate-adjusted posterior marginal mean predicted probabilities of 90-day modified Rankin Scale (mRS) scores by ICH hemisphere under the neutral prior. Percentages inside the bars represent posterior mean probabilities $\geq 3\%$. Panel B: Posterior density distributions of the predicted probabilities for each individual mRS category, stratified by left versus right hemisphere. Dotted lines indicate posterior medians. mRS = modified Rankin Scale. ICH = intracerebral hemorrhage.

Figure 2: modified Rankin Score at 90 days Stratified by ICH Laterality